

Virtual Sets

mark

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1 The Category of Virtual Sets

Definition 1.1 Given an injective map $f : X + A \rightarrow Y + A$, where A is finite, we define a new map

$$(f - A) : X \rightarrow Y$$

$$(f - A)(x) = \begin{cases} y & \text{if } f(x) = y : Y \\ f_A(a) & \text{if } f(x) = a : A \end{cases}$$

Where we define a helper for each $S \subseteq A$

$$f_S : S \rightarrow Y$$

$$f_S(a) = \begin{cases} y & \text{if } f(a) = y : Y \\ f_{S \setminus a}(a') & \text{if } f(a) = a' : A \end{cases}$$

Note by injectivity of f we have that a and a' are different, and since A , we will eventually hit an element of Y .

Theorem 1.2 *The operation $(_ - A)$ forms a trace on the category of finite sites with injections. That is:*

1. For any $f : Y + A \rightarrow Z + A$ and $g : X \rightarrow Y$ we have:

$$(f \circ (g + A)) - A = (f - A) \circ g$$

2. For any $f : Y \rightarrow Z$ and $g : X + A \rightarrow Y + A$ we have:

$$((f + A) \circ g) - A = f \circ (g - A)$$

3. For any $f : X + A \rightarrow Y + B$ and $g : B \rightarrow A$

$$((Y + g) \circ f) - A = (f \circ (X + g)) - B$$

4. For any $f : X + 0 \rightarrow Y \oplus 0$:

$$f - 0 = \rho_Y \circ f \circ \rho_X^{-1}$$

where $\rho_Z : Z \rightarrow Z + 0$ is the natural isomorphism.

5. For any $f : X + A + B \rightarrow Y + A + B$:

$$(f - B) - A = f - (A + B)$$

6. For any $f : X + A \rightarrow Y + A$ and $g : W \rightarrow Z$

$$g + (f - A) = (g + f) - A$$

Proof. We present a partial proof in the following lemmas. □

Lemma 1.3 Given $f : X \rightarrow Y$ we have $(f + A) - A = f$.

Proof. For any $x : X$ we have $(f + A)(\text{inl } x) = \text{inl } f(x) : Y$. Hence we see by definition that $((f + A) - A)(x) = f(x)$. □

Lemma 1.4 Given $f : X + A \rightarrow Y + A$ with A finite, for any set B we have

$$(f - A) + B = (f + B) - A$$

Proof. Note for any $b : B$ we have $((f + B) - A)(b) = b$ by definition. and for $x : X$ we have $(f + B)(x) = f(x)$, and similarly for any $a : A$. Unwinding definitions makes it clear that f_A and $(f + B)_A$ are the same function, and further that $((f + B) - A)(x) = (f - A)(x)$. □

Lemma 1.5 Given $f : X + A + B \rightarrow Y + A + B$ we have $(f - B) - A = f - (A + B)$.

Proof. Need to come up with a better way of seeing this but we can redefine the notion of subtraction in the following way: For an easier way to see this think of a function $X + A \rightarrow Y + A$ as a kind of graph on the vertex set $X + A + Y + B$, with edges $x \rightarrow f(x)$. Then we can take a graph quotient identifying the two copies of A and the left copy of A with its image under f . So $\text{inl } a \sim \text{inr } a$ and $\text{inl } a \sim f(a)$. Taking this graph quotient results in the graph of the function $f - A : X \rightarrow Y$. This is a kind of “small step” reduction process to produce $f - A$. Then we can see clearly that $f - B - A = f - (A + B)$ by just checking the two equivalence relations are the same, which they clearly are. □

Lemma 1.6 We have $(g + A) \circ f - A = g \circ (f - A)$. Similarly $g \circ (f + A) = (g - A) \circ f$.

Given these we are now in a position to work with virtual sets.

Definition 1.7 Define a category by

- Objects pairs of finite sets (X^+, X^-)
- Morphisms $X \rightarrow Y$ given by an injective map $X^+ + Y^- \rightarrow Y^+ + X^-$.
- Identities given by identity maps
- Composition given by the following construction. Given $f : X^+ + Y^- \rightarrow Y^+ + X^-$ and $g : Y^+ + Z^- \rightarrow Z^+ + Y^-$ we define a map

$$\begin{aligned} h : X^+ + Z^- + Y^- &\xrightarrow{f+Z^-} Y^+ + Z^- + X^- \\ &\xrightarrow{g+X^-} Z^+ + Y^- + X^- \\ &\xrightarrow{\sim} Z^+ + X^- + Y^- \end{aligned}$$

We will continue to abuse coherences and just write h as $(g + X^-) \circ (f + Z^-)$. The composition is given by $g \circ f := h - Y^-$.

We now check that composition is associative.

Proposition 1.8 *Composition is associative.*

Proof. Let $f : A \rightarrow B$, $g : B \rightarrow C$ and $h : C \rightarrow D$. Then we calculate using the lemmas before

$$g \circ f = (g + A^-) \circ (f + C^-) - B^-$$

$$\begin{aligned} h \circ (g \circ f) &= (h + A^-) \circ (g \circ f + D^-) - C^- \\ &= (h + A^-) \circ (((g + A^-) \circ (f + C^-) - B^-) + D^-) - C^- \\ &= (h + A^-) \circ (((g + A^-) \circ (f + C^-) + D^-) - B^-) - C^- \\ &= (h + A^-) \circ ((g + A^- + D^-) \circ (f + C^- + D^-) - B^-) - C^- \\ &= (h + A^- + B^-) \circ ((g + A^- + D^-) \circ (f + C^- + D^-)) - B^- - C^- \\ &= (h + A^- + B^-) \circ (g + A^- + D^-) \circ (f + C^- + D^-) - (C^- + B^-) \end{aligned}$$

Which is precisely the natural “triple composition” given between three morphisms.

Similarly we have

$$h \circ g = (h + B^-) \circ (g + D^-) - C^-$$

$$\begin{aligned} (h \circ g) \circ f &= (h \circ g + A^-) \circ (f + D^-) - B^- \\ &= (((h + B^-) \circ (g + D^-) - C^-) + A^-) \circ (f + D^-) - B^- \\ &= (((h + B^-) \circ (g + D^-) + A^-) - C^-) \circ (f + D^-) - B^- \\ &= ((h + B^- + A^-) \circ (g + D^- + A^-) - C^-) \circ (f + D^-) - B^- \\ &= ((h + B^- + A^-) \circ (g + D^- + A^-)) \circ (f + D^- + C^-) - C^- - B^- \\ &= (h + B^- + A^-) \circ (g + D^- + A^-) \circ (f + D^- + C^-) - (B^- + C^-) \\ &= (h + B^- + A^-) \circ (g + D^- + A^-) \circ (f + D^- + C^-) - (C^- + B^-) \quad \square \end{aligned}$$

Remark 1.9 Here is a relatively convincing reason why this category might not be correct: two objects are only isomorphic if their pairs of objects are isomorphic. This is unlike the intuition that we would expect $(3, 2) \simeq (2, 1)$.

We prove this now. Suppose $X \simeq Y$ in the category. Suppose they don't have the same size (this is fine because of decidability). WLoG suppose $|Y^+| < |X^+|$. Call $f : X \rightarrow Y$ and $g : Y \rightarrow X$ the functions in the isomorphism. Then note that for some $x^+ : X^+$ we must have $f(x^+) \in X^-$, due to size. Then by definition $(g + X^-)((f + X^-)(x^+)) = f(x^+)$, since g acts constantly on this copy of X^- . Then clearly $(g \circ f)(x^+) = f(x^+) \neq x^+$. So we must have $|Y^+| \geq |X^+|$. Arguing dually we see that $|Y^+| = |X^+|$ and so we deduce $|Y^-| = |X^-|$.

Composition behaves differently then we would expect on most morphisms. But on morphisms given by a pair of injective maps everything works out correctly.

Definition 1.10 Specifically given $f : X^+ \rightarrow Y^+$ and $g : Y^- \rightarrow X^-$ we obtain a map

$$\langle f, g \rangle := f + g : X^+ + Y^- \rightarrow Y^+ + X^- = X \rightarrow Y$$

Lemma 1.11 *Given $\langle f_0, g_0 \rangle : X \rightarrow Y$ and $\langle f_1, g_1 \rangle : Y \rightarrow Z$ we have*

$$\langle f_1, g_1 \rangle \circ \langle f_0, g_0 \rangle = \langle f_1 \circ f_0, g_1 \circ g_0 \rangle$$

Proof. This is clear since $(f_1 + g_1 + X^-) \circ (f_0 + g_0 + Z^-) = (f_1 \circ f_0) + g_1 + g_0 : X^+ + Z^- + Y^- \rightarrow Z^+ + Y^- + X^-$ Thus subtracting off ... **TODO:** $\square$

Let W be the class of morphisms which are bijections as set functions.

Definition 1.12 A **congruence** on a category $\mathcal{C}$ is an equivalence relation $\sim_{a,b}$ on each $\text{hom}_{\mathcal{C}}(a, b)$ such that if $f_1 \sim_{a,b} f_2$ and $g_1 \sim_{b,c} g_2$ then $g_1 \circ f_1 \sim_{a,c} g_2 \circ f_2$.

Given a family of relations $R_{a,b}$ on $\text{hom}_{\mathcal{C}}(a, b)$ we can create the congruence generated by R by taking the symmetric, reflexive, transitive and composition closure of R .

Let $\sim$ be the congruence relation on $\mathcal{C}$ generated by for all $f \in W$ that $f \circ f^{-1} \sim \text{id}$.

We define the category $\text{VSet} := \mathcal{C} / \sim$ to be the quotient of $\mathcal{C}$ by this congruence relation.

Definition 1.13 The category $\mathcal{C}$ has two monoidal products on it, given by addition and multiplication. Specifically

$$\oplus : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$$

$$X \oplus Y := (X^+ + Y^+, X^- + Y^-)$$

Functorial action is given by identity on the added component The associator is given by the isomorphism on $\mathcal{C}$ by the pair of associators on each component. The monoidal unit is given by $(0, 0) \in \mathcal{C}$, with similar unitors given by pairs of unitors.

Since composition of pairs acts just as composition pairwise, the coherence conditions reduce to coherence conditions in Set .

The second monoidal product is given by

$$X \otimes Y := (X^+ \times Y^+ + X^- \times Y^-, X^+ \times Y^- + X^- \times Y^+)$$

With unit given by $(1, 0)$.

Theorem 1.14 *These monoidal products descend to the category VSet with the quotient $\mathcal{C} \rightarrow \text{VSet}$ being a monoidal functor.*